

Question:

A 16-year-old boy presents with shortness of breath after prolonged exposure to cold air during a recent hike with his friends. Past medical history is significant for asthma, untreated because he doesn't like using medications. The patient says he is a non-smoker and occasionally drinks alcohol. On physical examination, his temperature is 37.0°C (98.6°F), pulse rate is 120/min, blood pressure is 114/76 mm Hg, and respiratory rate is 32/min. Auscultation of the chest reveals bilateral wheezing. Nebulized ipratropium bromide results in significant clinical improvement. Which of the following second messenger systems is affected by this drug? Answer Choices: A. Cyclic guanosine monophosphate (cGMP) system B. Arachidonic acid system C. Phosphoinositol system D. Tyrosine kinase system

Reasoning + Answer:

<think> To approach this reasoning problem, we need to analyze the context of the question, which is centered around identifying the second messenger system affected by ipratropium bromide, a drug commonly used in asthma management. The correct answer is derived from the understanding that ipratropium bromide acts on the muscarinic receptors, which is part of the phosphoinositol system in the context of bronchodilation. This leads us to select "Phosphoinositol system" as the most appropriate and accurate answer from the given options. </think> <answer> Phosphoinositol system </answer>

